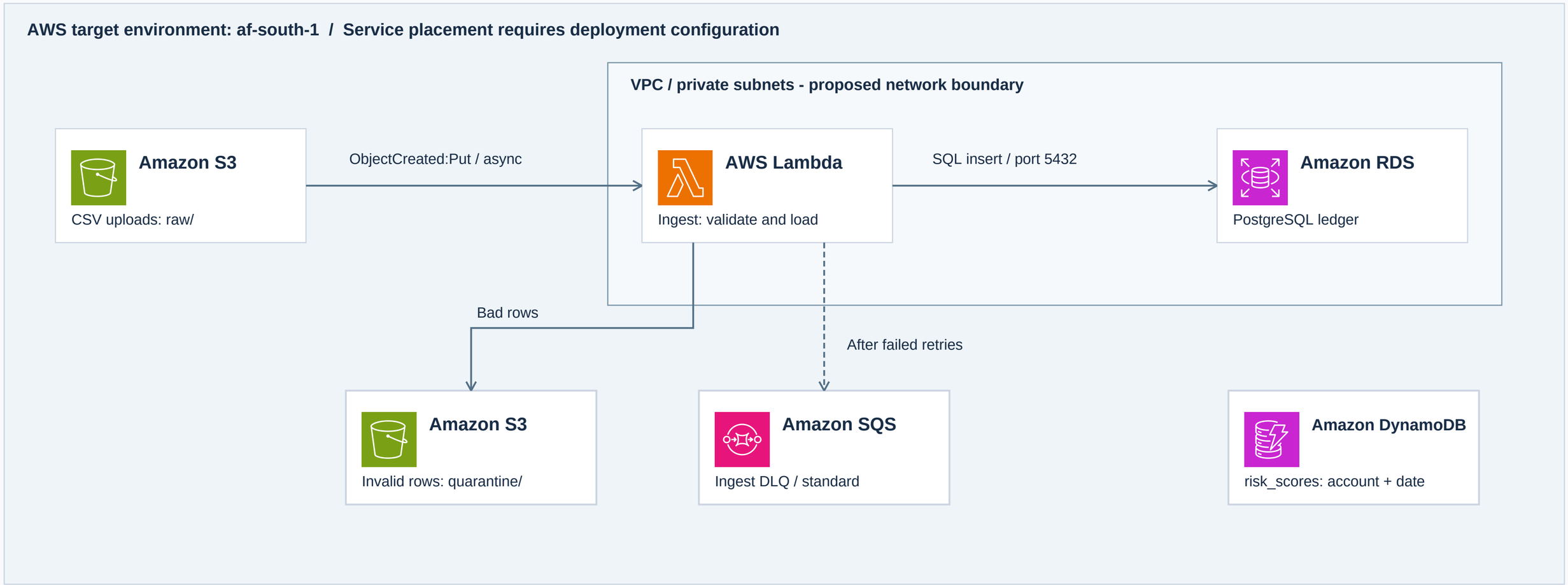
01 / Day 1-3 implementation

Application flow and provisioning scripts in the repository. AWS deployment has not been verified.



READ API

Flask / local + Lambda adapter

/v1/health and /v1/transactions -> PostgreSQL

/v1/insights -> DynamoDB (preloaded scores)

OPERATIONS AND SECURITY

Scripts: IAM roles, SSM parameters, CloudWatch alarms

Missing: KMS setup, access logging, VPC deployment

No automated scoring, email delivery or monthly reports yet

02 / Final capstone target

Planned architecture. Replace the direct ingest-to-RDS write when introducing FIFO; this is not a deployed-system claim.

AWS / af-south-1 | IAM per function | Customer-managed KMS keys | CloudWatch logs and alarms

Amazon S3
raw/ and quarantine/

AWS Lambda
Ingest / validation

Amazon SQS
FIFO / group by account

VPC / private subnets

AWS Lambda
Scoring + persistence

Amazon RDS
PostgreSQL

Query reads ledger; VPC attachment required

API Gateway
IAM-authorized data routes

AWS Lambda
Query / read API

Amazon DynamoDB
Daily account risk scores

AWS Lambda
Alert dispatcher

Amazon SES
Email alerts

EventBridge
Monthly schedule

AWS Lambda
Report / query ledger

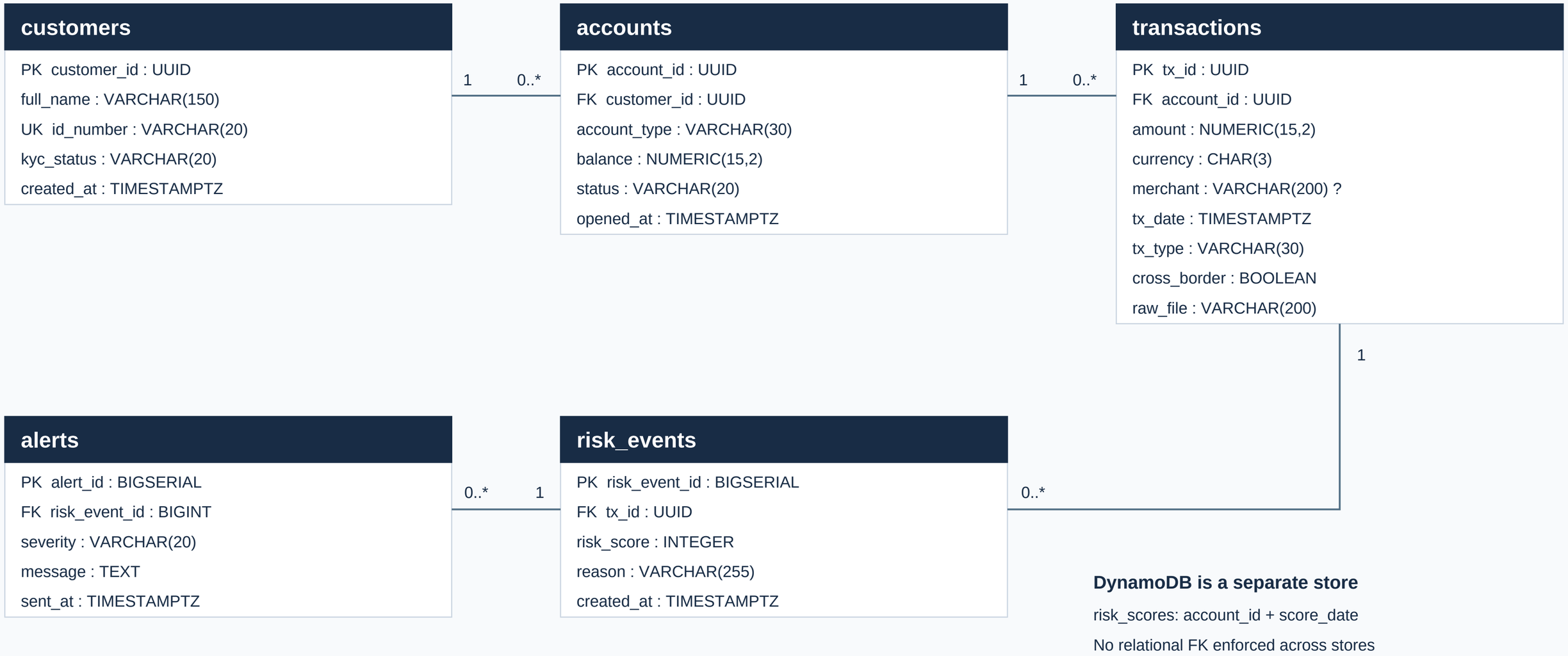
Amazon S3
CSV and JSON reports

RESILIENCE / DEPLOYMENT GAPS

- Separate ingest and scoring DLQs; replay tests
- Private AWS service access must be designed
- Budget review: no NAT; endpoints and KMS have costs
- Nightly Step Functions workflow: separate later scope

03 / PostgreSQL entity relationship diagram

Schema: sql/schema.sql | All attributes shown | PK primary key, FK foreign key, UK unique | ? nullable; others NOT NULL



Constraints: amount > 0; risk_score 0-100; KYC, account type, currency and transaction type allowlists.

Customer/account deletes are restricted. Each child has exactly one parent; a parent may have zero or many children.